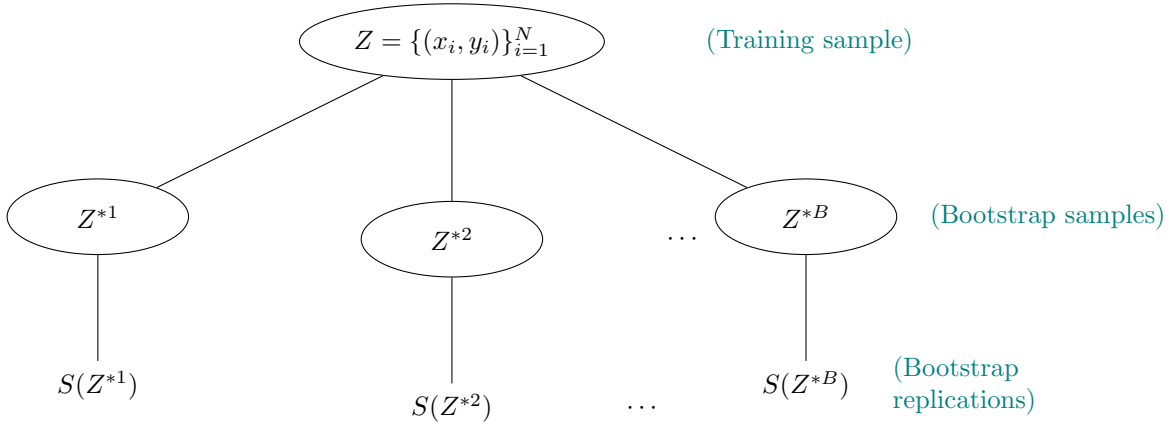


Bootstrap methods

- Bootstrap is a general tool for assessing statistical accuracy that can be used to estimate extra-sample prediction error.
- Let $Z = \{(x_i, y_i)\}_{i=1}^N$ be the training set. The basic idea is to randomly draw datasets with replacement from the training data, each sample the same size as the original training set, B times.

Suppose we wish to assess the statistical accuracy of the quantity $S(Z)$ computed from the dataset:



- From the bootstrap sampling, we can estimate any aspect of the distribution of $S(Z)$, e.g.:

$$\hat{\text{Var}}[S(Z)] = \frac{1}{B-1} \sum_{b=1}^B \left(S(Z^{*b}) - \bar{S}^* \right)^2, \quad (1)$$

where $\bar{S}^* = \sum_{b=1}^B S(Z^{*b})/B$.

An idea to estimate prediction error would be to fit the model on a set of bootstrap samples and keep track of how well it performs on the original training set. The estimate being:

$$\hat{\text{Err}}_{\text{Boot}} = \frac{1}{BN} \sum_{b=1}^B \sum_{i=1}^N L(y_i, \hat{f}^{*b}(x_i)) \quad (2)$$

- The problem with this approach is that the bootstrap datasets are acting as the training samples, while the original training set is acting as the test sample. But these two samples have observations in common.
- Actually, when drawing at random from Z , the probability that observation i is not picked in bootstrap sample is $(1 - \frac{1}{N})^N$, s.t.:

$$\mathbb{P}(i \in Z^{*b}) = 1 - \left(1 - \frac{1}{N}\right)^N \quad (3)$$

$$= 1 - e^{-1} \underset{(N \rightarrow \infty)}{\simeq} 0.632, \quad \text{hence} \quad (4)$$

for N large, even though we draw N samples, we only capture about 63.2% of the original distinct observations in any bootstrap sample.

- This provides the intuition for the Leave-One-Out Bootstrap, by using the out-of-bag observations (the 36.8% remaining):

$$\hat{\text{Err}}^{(1)} = \frac{1}{N} \sum_{i=1}^N \frac{1}{|C^{-i}|} \sum_{b \in C^{-i}} L(y_i, \hat{f}^{*b}(x_i)) \quad (5)$$

where C^{-i} is the set of indices b for bootstrap sample that do not contain observation i . Hence $\hat{\text{Err}}^{(1)}$ calculates the average prediction error only using models that were trained without seeing the observation they try to predict.

- One has to choose B large enough to ensure $|C^{-i}| > 0 \forall i$, or leaving the corresponding terms to zero.